Full support and positive density in the zero-temperature SK model

A reconstruction of OpenAI's quantitative argument · 25 October 2026

Abstract. Every integrable minimiser of the zero-temperature Parisi functional for the pure, zero-field Sherrington–Kirkpatrick model has full Stieltjes support on $[0,1)$ and a smooth density that is strictly positive on $(0,1)$. We give a short account of OpenAI's quantitative argument, organised around two uses of one spatial-moment inequality. On a positive constant interval, it makes the derivative of the gradient second moment strictly convex, contradicting the endpoint conditions imposed by minimality. A separate boundary estimate excludes terminal gaps. Full support then gives moment identities and smoothness; a local constant replacement transfers the inequality to the minimiser and forces its density to be positive. Chen previously proved full support and smooth nonnegative density. We explain the construction and uses of the quantitative estimate while importing its shape comparisons and full polynomial identity. We also derive the scalar value identity and state the classical optimisation consequence and the additional steps leading to asymptotically optimal quantum approximate optimization algorithm (QAOA) circuits.

1. The result and the mechanism

For independent standard Gaussian couplings J_{ij}, the pure, zero-field Sherrington–Kirkpatrick (SK) Hamiltonian is

$$H_n(\sigma) = n^{-1/2} \sum_{i<j} J_{ij}\sigma_i\sigma_j, \qquad \sigma \in \{-1,1\}^n.$$

The zero-temperature problem concerns the limiting maximum energy per spin,

$$P_* := \lim_{n\to\infty} n^{-1}\mathbb{E}\max_{\sigma} H_n(\sigma).$$

The Parisi formula expresses P_* as a minimum over nondecreasing functions. Our subject is the structure of a minimiser: its increments charge every interval of overlap scales below one, and in fact have strictly positive density at each interior scale. The proof uses one quantitative bound twice, first to exclude constant intervals and then, after an approximation, to prove pointwise positivity.

For two configurations, their overlap is $R(\sigma,\tau) = n^{-1}\sum_i \sigma_i\tau_i$. It determines the energy covariance: $\mathbb{E}[H_n(\sigma)H_n(\tau)] = nR^2/2 - 1/2$. Thus the Gaussian law of the energies is encoded by the overlaps. We use the covariance convention $\xi(t) = t^2/2$, so $\xi'' = 1$; adding one independent centred Gaussian of variance $1/2$ to every configuration's energy gives covariance $n\xi(R)$ and leaves the expected maximum unchanged. This is the normalisation of the ground-state Parisi formula of Auffinger and Chen (2017), Theorem 1. There are no higher-order interactions and no external field.

Let $\mathcal{U}$ consist of the nonnegative, nondecreasing, right-continuous functions $\gamma : [0,1) \to [0,\infty)$ satisfying $\int_0^1 \gamma(t)\,dt < \infty$. Define the locally finite Stieltjes measure by

$$\mu_\gamma([0,t]) = \gamma(t), \qquad 0 \le t < 1.$$

Thus $\mu_\gamma((s,t]) = \gamma(t) - \gamma(s)$: the measure records the increments of γ. A point mass (atom) of size $\gamma(0)$ is allowed at zero, and the total mass need not be finite or equal to one. All support statements use the relative topology of $[0,1)$. A support gap is a relatively open interval of positive length with zero mass; the cumulative coefficient is constant on such an interval.

The Parisi functional $\mathcal{P}(\gamma)$ is defined using an auxiliary value function $\Phi_\gamma(t,x)$, where $x \in \mathbb{R}$ is a spatial variable and t is the overlap parameter, treated as time in the equation. Write $\Phi = \Phi_\gamma$ and

use subscripts for partial derivatives. The backward equation, terminal condition and functional are

$$\Phi_t = -\tfrac{1}{2}(\Phi_{xx} + \gamma(t)\Phi_x^2), \qquad \Phi(1,x) = |x|, \qquad \mathcal{P}(\gamma) = \Phi_\gamma(0,0) - \tfrac{1}{2}\int_0^1 t\gamma(t)\,dt.$$

The terminal value $|x|$ is the zero-temperature value of one spin in a field x. For an integrable coefficient, the PDE solution is interpreted by approximation from bounded monotone coefficients. The exact control definition and stability theorem are OpenAI (2026a), §1 and Lemmas 2.1–2.2. To turn minimality of $\mathcal{P}$ into a condition on γ, we use a random process whose gradient moment measures the first-order cost of changing γ. Write $u = \Phi_x$ for the spatial gradient and $a = \Phi_{xx}$ for its spatial derivative. Let B be standard Brownian motion and define the diffusion X and the moment q by

$$dX_t = \gamma(t)u(t,X_t)\,dt + dB_t, \qquad X_0 = 0, \qquad q(t) = \mathbb{E}[u(t,X_t)^2].$$

The discrepancy $q(t) - t$ enters the first variation of $\mathcal{P}$; minimality will force $q(t) = t$ on the support of μ_γ. The diffusion exists uniquely for the fixed Brownian motion. All stochastic calculations below take place on compact time intervals ending before one, unless an endpoint limit is justified explicitly.

Auffinger, Chen and Zeng (2020), Theorem 1 proved infinite support. This need not be full support: the measure $\sum_{k\geq 1} 2^{-k}\delta_{1-2^{-k}}$ has infinitely many atoms and many gaps. Chen (2026), Theorem 1.3 subsequently proved full support and smooth nonnegative density. We follow OpenAI's quantitative proof of the support conclusion and its further strict-positivity result. The feature used twice is a lower bound valid throughout every positive constant interval of an admissible coefficient, without a minimising assumption.

Theorem 1 (full support; OpenAI (2026a), Theorem 1.1). If $\gamma \in \mathcal{U}$ minimises $\mathcal{P}$, then

$$\operatorname{supp}\mu_\gamma = [0,1), \qquad \gamma(t_2) > \gamma(t_1) \quad (0 \leq t_1 < t_2 < 1).$$

For every associated diffusion and every $0 \leq t < 1$,

$$q(t) = t, \qquad \mathbb{E}[a(t,X_t)^2] = 1.$$

Existence of a minimiser is a separate input from Auffinger and Chen (2017), Theorem 1 and the proof of Lemma 3. Integrability allows γ to grow near one. Zero field gives $u(0,0) = 0$, and $\xi'' = 1$ is used in the curvature calculation; the theorem here does not extend these hypotheses to nonzero field or a general mixed-spin covariance.

The proof compares two demands on a hypothetical gap. Minimality forces q to agree with time at its endpoints, with endpoint slopes at most one. On a positive constant interval a quantitative estimate makes q' strictly convex, so its integral is smaller than the interval's length. This contradicts the required change in q. After excluding all gaps, the identity $q' = 1$ supplies the equation from which the same estimate will force $\gamma' > 0$. This second use needs the estimate for a local constant replacement that need not minimise, without requiring $q'' = 0$ at the comparison time. Its explicit positive margin must also survive the replacement's limit.

The support identities also supply the normalisation used in a proof of asymptotic optimality for the quantum approximate optimization algorithm (QAOA). Section 5 states that separate theorem and the finite Gaussian correlation quantity it uses. The density conclusion is not needed for that

construction or for the classical consequence stated there. We assume Itô calculus and martingale convergence. Omitted proofs are linked to *From Parisi full support to deterministic QAOA circuits: A long reasoning record*, the companion edition of 25 October 2026; original-source citations remain beside the imported inputs.

2. Minimality gives contact; curvature excludes gaps

For an admissible coefficient, Φ is even, convex and one-Lipschitz in space. Its positive-order spatial derivatives are bounded and continuous on each strip $[0,T]\times\mathbb{R}$, $T<1$; the PDE holds almost everywhere in time. The compact-strip approximation theorem justifies the following Itô calculations even when γ has jumps (see OpenAI (2026a), Lemmas 2.1–2.2). The control method has antecedents in Jagannath and Tobasco (2016) at positive temperature and Auffinger and Chen (2017), Theorem 2 and Corollary 2 at zero temperature; see also Chen, Handschy and Lerman (2018), Proposition 2 and Theorem 5.

Fix a minimiser. Itô's formula cancels the drift of $u(t,X_t)$:

$$du(t,X_t)=a(t,X_t)\,dB_t,\qquad q(0)=0,\qquad q'(t)=\mathbb{E}[a(t,X_t)^2].$$

In particular q' is continuous across coefficient jumps. Minimality means that every allowed one-sided change has nonnegative first-order cost. The one-sided variation formula gives the rate of change of the functional when γ is replaced by $\gamma+\varepsilon\eta$ as $\varepsilon\downarrow 0$:

$$D_+\mathcal{P}(\gamma)[\eta]=\tfrac12\int_0^1\eta(t)(q(t)-t)\,dt, \tag{1}$$

for every integrable direction η for which $\gamma+\varepsilon\eta\in\mathcal{U}$ for all sufficiently small $\varepsilon\ge 0$ (see OpenAI (2026a), Lemma 6.1). Earlier zero-temperature variation and support identities appear in Chen, Handschy and Lerman (2018), Propositions 3–4.

Adding a unit atom at s raises the cumulative function at every $t\ge s$, so its direction is $\eta=\mathbf{1}_{[s,1)}$. The first-order cost is therefore a tail integral:

$$G(s):=\int_s^1(q(t)-t)\,dt\ge 0.$$

Here $G(s)/2$ is the first-order cost of the added atom. Conversely, let $\nu=\mu_\gamma|_K$ for a compact $K\subset[0,1)$. For $0\le\varepsilon\le 1$, the measure $\mu_\gamma-\varepsilon\nu=(1-\varepsilon\mathbf{1}_K)\mu_\gamma$ is nonnegative, and its cumulative function is $\gamma-\varepsilon\nu([0,\cdot])\in\mathcal{U}$. Minimality gives

$$0\le D_+\mathcal{P}(\gamma)[-\nu([0,\cdot])]=-\tfrac12\int_K G(s)\,d\mu_\gamma(s).$$

The change of integration order is justified by $|q-t|\le 1$ and $\mu_\gamma(K)<\infty$. Since $G\ge 0$, it vanishes almost everywhere for μ_γ, and continuity makes it vanish on $S=\operatorname{supp}\mu_\gamma$. The variation removes a fraction of all mass in K, not only atoms; no density or probability constraint is assumed.

At a support point $s>0$, G has a minimum. Hence

$$G'(s)=s-q(s)=0,\qquad G''(s)=1-q'(s)\ge 0,\quad\text{so}\quad q(s)=s,\quad q'(s)\le 1. \tag{2}$$

If $0\in S$, contact gives $G(0)=0$; independently, zero field gives $q(0)=0$. Hence $G(h)=\frac12(1-q'(0))h^2+o(h^2)$; nonnegativity gives the same slope bound at zero. These are the contact conditions of OpenAI (2026a), Lemma 6.2.

Lemma 2 (constant-interval curvature; OpenAI (2026a), Proposition 5.1). Let $\gamma \in \mathcal{U}$, with its associated diffusion and moment q. If $\gamma = c$ on an open interval $I \subset (0,1)$, called a plateau, then for $t \in I$

$$q'''(t) \geq \frac{c^2}{1000}\mathbb{E}[a(t, X_t)^4] > 0 \quad \text{if } c > 0. \tag{3}$$

If $c = 0$, then $q''(t) = \mathbb{E}[\Phi_{xxx}(t, X_t)^2] > 0$.

Minimality is not a hypothesis of this lemma. Section 3 derives its moment polynomial and explains the integrated estimate imported from the source. We first use the lemma to exclude gaps. A complementary interval with two support endpoints admits two contact conditions; an initial interval without support at zero has coefficient zero; a terminal interval requires a boundary estimate.

Suppose (ℓ, r) is a support gap with endpoints in S and $r < 1$. The coefficient is constant there. If its value is positive, strict convexity of q' and (2) give

$$q'(t) < \frac{r-t}{r-\ell}q'(\ell) + \frac{t-\ell}{r-\ell}q'(r) \leq 1 \qquad (\ell < t < r).$$

Integrating contradicts $q(r) - q(\ell) = r - \ell$. The geometric obstruction is an area deficit under q'. If the constant is zero, strict increase gives $q'(t) < q'(r) \leq 1$ and the same contradiction. An initial gap $[0, r)$ with $0 \notin S$ and $r \in S$ has no mass, hence $\gamma = 0$ there. Its right endpoint gives $q(r) = r$ and $q'(r) \leq 1$, while zero field gives $q(0) = 0$. The same integral contradiction follows without assuming $G(0) = 0$. A gap $(0, r)$ with $0 \in S$ is already covered by the two-contact case.

A terminal gap cannot use contact at one, which lies outside the support domain. Its exclusion was recorded by Auffinger, Chen and Zeng (2020), Remark 7. Chen (2026), Proposition 4.5, equations (4.27)–(4.38) proves a positive square-root asymptotic for $1-q(t)$ under the terminal-gap hypothesis. We use the sufficient lower bound proved directly in OpenAI (2026a), Lemma 6.3: on a supposed constant interval ending at one,

$$1 - q(t) \geq C\sqrt{1-t}$$

for some $C > 0$ and all t sufficiently close to one. Two estimates produce this scale. The heat representation of the terminal value $|x|$ gives $1 - u(t,x)^2 \geq \kappa > 0$ on $|x| \leq \sqrt{1-t}$. A transition-kernel comparison from a fixed earlier time gives a density $f_t(x) \geq m > 0$ on the fixed window $|x| \leq 1$, uniformly as $t \uparrow 1$. Thus

$$1 - q(t) = \int_{\mathbb{R}} (1 - u(t,x)^2) f_t(x)\, dx \geq 2m\kappa\sqrt{1-t}.$$

The time-uniform density estimate, proved in Lemma 6.3, equations (6.6)–(6.7), is essential; pointwise positivity of each density would not suffice. For $h = 1-t < C^2$, we have $q(t)-t \leq h - C\sqrt{h} < 0$, so $G(s) < 0$ for s sufficiently close to one. This contradicts minimality and also excludes empty support. The zero-coefficient heat argument and late-step variation have antecedents in Chen, Handschy and Lerman (2018), Theorem 6 and its proof.

These cases exhaust the complement of the relatively closed set S. Consequently $S = [0, 1)$, contact gives $q(t) = t$, and differentiation gives $\mathbb{E}a^2 = 1$, including the right-hand value at zero. Moreover, $\gamma(t_2) - \gamma(t_1) = \mu_\gamma((t_1, t_2]) > 0$. This proves Theorem 1 using the stated analytic inputs.

3. The integrated estimate

The quantity used in both arguments is a spatial moment. Write $b = \Phi_{xxx}$ and $d = \Phi_{xxxx}$, and, for $0 < t < 1$, set

$$\mathcal{C}_{\gamma}(t) = \mathbb{E}[d^2 - 12\gamma(t)ab^2 + 6\gamma(t)^2a^4],$$

with the spatial factors evaluated at (t, X_t). Its definition requires spatial regularity but no time derivative of γ. On a plateau it equals q'''; after full support and smoothness, it will instead determine γ'.

On a plateau with $\gamma = c > 0$, put $r = c\Phi_x = cu$, $v = r_x = ca$, $w = r_{xx}$ and $z = r_{xxx}$. This rescaling makes the derivative identities polynomial. The PDE becomes $r_t = -\frac{1}{2}r_{xx} - rr_x$, and the diffusion has spatial generator $\frac{1}{2}\partial_{xx} + r\partial_x$. Differentiating in space and applying Itô's formula gives the drift and diffusion pairs $(0, v)$, $(-v^2, w)$ and $(-3vw, z)$ for r, v, w, respectively. All expectations in this section evaluate spatial functions at (t, X_t). Starting with $c^2q' = \mathbb{E}v^2$, Itô's formula gives

$$\begin{aligned} c^2q'' &= \mathbb{E}[w^2 - 2v^3], \\ c^2q''' &= \mathbb{E}[z^2 - 12vw^2 + 6v^4] = c^2\mathcal{C}_{\gamma}(t). \end{aligned} \tag{4}$$

For example, the drifts of w^2 and v^3 are $z^2 - 6vw^2$ and $3vw^2 - 3v^4$, respectively. The negative mixed term is the obstacle. Convexity, which gives $v \geq 0$, does not control it. Indeed, for the smooth auxiliary terminal value $\log(2\cosh x)$ and coefficient $\gamma \equiv 1$, the solution is $\Phi(t, x) = \log(2\cosh x) + (1 - t)/2$. Thus $r = \tanh x$; at $r^2 = 1/2$, the polynomial in (4) equals $-19/8$. This is a finite-step smoothing example, not a minimiser of the absolute-value-terminal problem; its calculation is given in long §4.1.

The proof subtracts a spatial derivative whose expectation under the law of X_t is zero. The resulting expression has a positive fourth-moment remainder. The backward value function and the forward density must satisfy joint shape restrictions for this decomposition to have the required signs. Those comparisons and the exact polynomial identity are the substantial inputs from OpenAI (2026a), §§3–4. We explain their roles first for finite-step coefficients, then pass the integrated inequality to an integrable coefficient by approximation.

Begin with a positive nondecreasing finite-step coefficient γ_{step}: a function constant on each interval of a finite partition. Replace the terminal value $|x|$ by the smooth approximation $M^{-1}\log(2\cosh Mx)$, with $M \geq \|\gamma_{\text{step}}\|_{\infty}$. On a step of height c, the exponential $e^{c\Phi}$ solves the backward heat equation. At positive time, write the diffusion density as $f = e^{c\Phi}g$. This substitution turns its Fokker–Planck equation into the forward heat equation for g. For $s = -\partial_x \log g$, the source proves joint shape restrictions: on $x > 0$, $r_x > 0$ and $s_x \geq 1/t$, both r and s are concave, and $H := w/(2r) + v \leq 0$ is nondecreasing. They also give

$$j := s_x - (s/r)v = r\,\partial_x(s/r) \geq 0.$$

The last inequality couples the backward gradient to the forward density; separate monotonicity and concavity do not supply it. For $\eta = j/s = s_x/s - v/r$ on $x > 0$, the comparison equation is

$$\eta_t = \tfrac{1}{2}\eta_{xx} + \frac{s_x}{s}\eta_x + (v/r)_x\eta + 2H_x - w - s_{xx}.$$

The source is nonnegative and $(v/r)_x < 0$. On the first step, $s = x/t$, so concavity of r gives $\eta = 1/x - v/r \geq 0$. Comparison preserves this sign within a step, while cancellation in the coefficient-jump rules gives $j_+ = j_-$ between steps. The quotients w/r and s/r extend smoothly

through zero because r, s, w are odd and $r_x(0) > 0$. The comparison's boundary controls and these extensions are analytic inputs from OpenAI (2026a), Lemmas 3.2–3.4, reconstructed in long §4.3.

The factorisation of the density gives $f_x/f = r - s$. For a suitable expression F in the shape quantities, define

$$\mathcal{W}(F) = r\partial_x F + (v + r^2 - rs)F.$$

Then $f\mathcal{W}(F) = \partial_x(rfF)$. The smooth extensions avoid an interior boundary term at zero, and the source's differentiated Gaussian tail estimates give $rfF \to 0$ at both spatial infinities for the polynomial correction used below. Hence $\mathbb{E}_f \mathcal{W}(F) = 0$, where $\mathbb{E}_f$ denotes integration against $f(x)\,dx$.

The remaining step is algebra under these analytic restrictions. OpenAI (2026a), Lemma 4.1 gives an explicit polynomial F_2 for which subtracting $\mathcal{W}(F_2)$ from $z^2 - 12vw^2 + 6v^4$ leaves a sum of weighted squares, nonnegative products and a polynomial with positive coefficients. The construction first arranges the terms involving the highest derivative z. In the source's polynomial variables, only differentiating $-w/r$ introduces z into the correction, through $r\partial_x(-w/r) = -z + wv/r$. The partial derivative of F_2 with respect to this variable is chosen to cancel the linear terms in z from the square and the nonnegative product. The remaining coefficients give the positive remainder; the complete choice is in the source's §4.1. For example, one square has the nonnegative weight $r^2 j$; this is where the joint comparison enters. The remainder's v^4 coefficient is $1661507/1521000000 > 1/1000$. Integration therefore gives

$$\int_{\mathbb{R}} (z^2 - 12vw^2 + 6v^4) f\,dx \geq \frac{1}{1000} \int_{\mathbb{R}} v^4 f\,dx. \tag{5}$$

We use (5) as an imported analytic estimate. The full polynomial and its decomposition are in Lemma 4.1 and Table 1; Proposition 4.2 integrates them using the signs and zero boundary flux supplied by Lemma 3.4. Long §4.4, "How the signs make a correction with zero expectation useful" reproduces the definitions and complete identity.

To obtain Lemma 2 for an integrable coefficient, choose positive monotone step coefficients γ_n retaining the plateau near a fixed interior time, with $\|\gamma_n - \gamma\|_1 \to 0$, and terminal parameters $M_n \geq \|\gamma_n\|_\infty$ with $M_n \to \infty$. Spatial derivatives through order four converge uniformly on a fixed strip below one, and the diffusions driven by the same Brownian motion converge (see OpenAI (2026a), Lemmas 2.1–2.2 and Proposition 5.1). Only v, w, z remain in (5), so no convergence theorem for the auxiliary logarithmic derivative s is needed. Dividing by c^2 gives (3); strictness follows from $\Phi_{xx} > 0$ on the plateau (see OpenAI (2026a), Lemma 2.3). The plateau-preserving approximants and their common strip bounds are specified in long §4.5. The zero plateau has the separate heat identity in Lemma 2. Its strictness uses a positive diffusion density and the fact that Φ_{xxx} cannot vanish identically: an even quadratic profile would contradict the one-Lipschitz bound and $|x| + O(1)$ growth.

The recent antecedents concern the particular shape and density comparisons used with the Cole–Hopf representation. Lopatto (2026), v3, Proposition 3.6 propagates backward shape inequalities through a finite cascade; Proposition 7.1 gives the transformed-density representation, and Proposition 9.1 gives a crossing argument at positive temperature. In the zero-temperature setting, Chen (2026), Proposition 4.2 proves $q''(t) = 0 \Rightarrow q'''(t) > 0$ on a hypothetical positive gap, under the minimiser convention of §3.1. Bound (3) instead holds throughout a positive plateau of any admissible coefficient and retains a quantitative fourth-moment term.

Nonminimiser scope alone is not the distinction: Lopatto's v4, Proposition 5.1 and Corollary 5.6, revised 5 October 2026, give a qualitative crossing theorem and a decrease-then-increase description for arbitrary positive-temperature probability measures. A related inequality has both support and density uses in Lopatto's external-field paper, Theorem 2.3 and Appendix B, under different temperature and one-sided coefficient hypotheses. Here the quantitative zero-temperature estimate must survive a local replacement that need not minimise or satisfy $q'' = 0$ at the chosen time. The edition-specific comparisons are in long §2.3.

4. A second use: the density cannot vanish inside the interval

Full support does not force pointwise positive density. For instance, $\gamma(t) = \int_0^t (s - \frac{1}{2})^2 \, ds$ has full Stieltjes support but $\gamma'(\frac{1}{2}) = 0$: every interval of positive length has a positive integral, although the integrand vanishes at one point. The quantitative remainder in (5) rules this out for a minimiser. First, the identity $\mathbb{E}a^2 = 1$ expresses γ as a quotient of continuous moments and then gives smoothness. Only afterwards do we transfer (5) from a locally constant comparison coefficient to obtain $\gamma' > 0$.

Corollary 3 (smooth positive density; OpenAI (2026a), Corollary 6.4). Under the hypotheses of Theorem 1, $\gamma(0) = 0$ and $\gamma \in C^\infty([0,1))$, with derivatives at zero taken from the right. Its Stieltjes measure is $\mu_\gamma(dt) = \gamma'(t)\, dt$, and

$$2\gamma'(t)\mathbb{E}[a(t, X_t)^3] \geq \frac{\gamma(t)^2}{1000}\mathbb{E}[a(t, X_t)^4] > 0 \qquad (0 < t < 1). \tag{6}$$

Only nonnegativity of the density at zero is asserted.

All spatial factors in expectations below are evaluated at (t, X_t). Since $da = -\gamma a^2\, dt + b\, dB$, Itô's formula applied to a^2 gives, for almost every t, $0 = \frac{d}{dt}\mathbb{E}a^2 = \mathbb{E}b^2 - 2\gamma\mathbb{E}a^3$. Here the left side vanishes because full support gives $\mathbb{E}a^2 = 1$. Solving for the coefficient yields

$$\gamma(t) = \frac{\mathbb{E}b^2}{2\mathbb{E}a^3}. \tag{7}$$

Initially (7) holds almost everywhere. Its denominator is at least two, because $a \geq 0$ and $\mathbb{E}a^3 \geq (\mathbb{E}a^2)^{3/2} = 1$. The moments are continuous; right continuity of γ extends (7) to every time and proves continuity of γ. Evenness gives $b(0,0) = 0$, hence $\gamma(0) = 0$.

For smoothness, fix $T < 1$ and consider all polynomial moments

$$M_P(t) = \mathbb{E}P(\Phi_x(t, X_t), \ldots, \partial_x^m \Phi(t, X_t)),$$

where P is a polynomial in any finite number m of variables. Spatial differentiation of the PDE and Itô's product rule give polynomials Q_P, R_P, involving finitely many further spatial derivatives, such that

$$M_P(t) - M_P(0) = \int_0^t [M_{Q_P}(s) + \gamma(s) M_{R_P}(s)]\, ds.$$

No derivative of γ enters this identity. Bounded spatial derivatives on the fixed strip make every moment continuous. The integral identity then makes all moments C^1, and (7) makes γ C^1. Repetition proves smoothness. This is the quotient-and-moment argument of Chen (2026), Proposition 4.6 and Appendix B.6, using the polynomial-moment method of Auffinger and Chen (2015), Lemma 2

and the proof of Theorem 2(ii). The explicit polynomials and the passage from initially nonsmooth coefficients are in OpenAI (2026a), proof of Corollary 6.4 and long §9.1.

Now the two differentiated moments are

$$\frac{d}{dt}\mathbb{E}b^2 = \mathbb{E}[d^2 - 6\gamma ab^2], \qquad \frac{d}{dt}\mathbb{E}a^3 = \mathbb{E}[3ab^2 - 3\gamma a^4].$$

For a smooth time coefficient, differentiating $q'' = \mathbb{E}b^2 - 2\gamma\mathbb{E}a^3$ therefore gives

$$q^{(3)} + 2\gamma'\mathbb{E}a^3 = \mathcal{C}_\gamma(t). \tag{8}$$

This identity displays the two uses of the same spatial moment. On a plateau, $\gamma' = 0$ and $(v, w, z) = c(a, b, d)$ recover (4). For the now smooth minimiser, $q = t$, so $q^{(3)} = 0$ and the left side is $2\gamma'\mathbb{E}a^3$. It remains to transfer the lower bound for $\mathcal{C}_\gamma(t)$ from a plateau to this minimiser. The identity separates the spatial moment that can converge from the time derivatives, whose limits need not agree.

Fix $t \in (0, 1)$ and $c = \gamma(t) > 0$. For $0 < \delta < \min(t, 1 - t)$ define

$$\gamma_\delta(s) = \begin{cases} c, & t - \delta \le s < t + \delta, \\ \gamma(s), & \text{otherwise.} \end{cases}$$

The inequalities $\gamma(t - \delta) \le c \le \gamma(t + \delta)$ make both joins nondecreasing. Thus $\gamma_\delta \in \mathcal{U}$; it need not minimise $\mathcal{P}$. Moreover,

$$\|\gamma_\delta - \gamma\|_1 \le 2\delta\,[\gamma(t + \delta) - \gamma(t - \delta)] \longrightarrow 0.$$

Use a subscript δ for its value-function derivatives and diffusion. Lemma 2 and (4), already extended to integrable coefficients, give at the fixed time t

$$\mathcal{C}_{\gamma_\delta}(t) = \mathbb{E}[d_\delta^2 - 12c\, a_\delta b_\delta^2 + 6c^2 a_\delta^4] \ge \frac{c^2}{1000}\mathbb{E}[a_\delta^4]. \tag{9}$$

Choose a single T with $t < T < 1$. Lemma 2.1 of OpenAI (2026a) gives uniform convergence in space and uniform bounds for $a_\delta, b_\delta, d_\delta$ on $[0, T]$, with the next spatial derivatives bounded as well. Lemma 2.2 couples the diffusions using the same Brownian motion and gives $\sup_{s \le T} |X_s^\delta - X_s| \to 0$. The derivative bounds control evaluation on these paths, so the integrands in (9) converge and are uniformly bounded. Dominated convergence passes their expectations to the limit. The common coefficient L^1 bound and the fixed distance from time one determine these estimates; the shrinking plateau width does not. The result is $\mathcal{C}_\gamma(t) \ge \gamma(t)^2\mathbb{E}a^4/1000$. This passage needs only an admissible coefficient continuous at t with $\gamma(t) > 0$; minimality enters when (8) gives (6). Strictness follows from $c > 0$ and $\mathbb{E}a^4 \ge (\mathbb{E}a^2)^2 = 1$. In particular, Cauchy–Schwarz gives

$$\mathbb{E}a^3 \le \sqrt{\mathbb{E}a^2\,\mathbb{E}a^4} = \sqrt{\mathbb{E}a^4} \le \mathbb{E}a^4,$$

so the same inequality implies $\gamma'(t) \ge \gamma(t)^2/2000$ for $0 < t < 1$.

We have passed spatial moments, not the time derivatives q_δ'''. Indeed the limiting minimiser has $q(t) = t$ and $q''' = 0$; the coefficient derivative in (8) is exactly what the constant-interval formula omits. Long §§9.1–9.2 retain the smoothness argument and fixed-time transfer, including the common approximation bounds.

5. Consequences for optimisation

5.1 The scalar value

The support identities give a scalar representation of P_*. Set $M_t = u(t, X_t)$, the bounded gradient martingale satisfying $dM_t = a(t, X_t)\, dB_t$. Full support gives $\mathbb{E}M_t^2 = t$ and $\mathbb{E}a(t, X_t)^2 = 1$: each increment $M_t - M_s$ has unconditional variance $t - s$ for $0 \le s < t < 1$. The second-moment identity will normalise the discrete increments below. The first moment of a instead measures the covariance gained with the driving Brownian motion, and recovers the value:

$$P_* = \mathbb{E}[UB_1] = \int_0^1 \mathbb{E}[a(t, X_t)]\, dt. \tag{10}$$

Here U is the almost-sure and L^2 limit of M_t. Boundedness gives $|U| \le 1$, while $\mathbb{E}M_t^2 \to 1$ gives $\mathbb{E}U^2 = 1$; therefore $U^2 = 1$ almost surely. To see the cancellation behind (10), the terminal PDE estimate gives $UX_1 = |X_1|$, while Itô's formula gives $\mathcal{P}(\gamma) = \mathbb{E}|X_1| - \int_0^1 t\gamma(t)\, dt$. Multiplying $X_1 = B_1 + \int_0^1 \gamma(t) M_t\, dt$ by U cancels that integral because $\mathbb{E}[UM_t] = t$. The Parisi formula then gives (10). The endpoint limits and stochastic-integral covariance are proved in OpenAI (2026a), Corollary 7.1 and the long record, §6, "Recovering the value from a martingale". This argument uses full support, not the density corollary.

The gradient-martingale and value methods have antecedents in Montanari, preprint v2, Lemmas 3.2–3.4, under its positive-temperature support assumption, and in the zero-temperature calculations of El Alaoui, Montanari and Sellke, preprint v1, Lemmas 6.5 and 6.13. Sellke, preprint v4, Proposition 3.10 and Lemma 3.11, p. 17 gives the closely related finite-cutoff cancellation by applying Itô's formula to Φ and $x\Phi_x$.

5.2 The classical algorithm

Full support and minimiser existence verify the no-overlap-gap assumption at zero temperature, Assumption 2 of El Alaoui, Montanari and Sellke, preprint v1. Their Corollary 2.2 then gives, for every fixed $\varepsilon > 0$, a randomised algorithm returning a configuration within εn of the finite SK maximum with probability tending to one over couplings and algorithmic randomness. Its cost is $C(\varepsilon)n^2$ real-arithmetic operations with the prescribed scalar functions and normalisers available; this includes no separate numerical-preprocessing or finite-precision bound. OpenAI (2026a), Corollary 7.3 and long §9.4 check the Gaussian convention and operation count. Chen already records the classical consequence in §1, p. 4. These identities enter the incremental approximate message-passing construction through the bounded martingale M_t. On a fixed cutoff, $\mathbb{E}a^2 = 1$ gives its variance $\mathbb{E}M_t^2 = t$; El Alaoui, Montanari and Sellke, Theorem 2, with the source's regularity and discretisation estimates, gives an energy lower bound arbitrarily close to the truncated first-moment integral in (10). Strict positivity of the density is unnecessary.

5.3 A finite QAOA word before the thermodynamic limit

QAOA alternates evolution under the cost Hamiltonian with transverse-field mixing, starting from the uniform superposition of spin configurations Farhi, Goldstone and Gutmann (2014). Each pair of operations is a layer, and their durations are its angles. For every $\varepsilon > 0$, OpenAI (2026b), Theorem 1.1 constructs a finite depth and deterministic angles, independent of n and the couplings J_{ij}, whose limiting disorder-averaged expected energy per spin is at least $P_* - \varepsilon$. The depth and angles may depend on ε; $n \to \infty$ is taken with them fixed. The earlier fixed-instance depth

limit in Farhi, Goldstone and Gutmann, §VI allows choices depending on the instance and does not supply this common finite word before the size limit. Farhi, Goldstone, Gutmann and Zhou (2022) established the fixed-angle thermodynamic limits in this convention. The eventual-optimality conclusion was conjectured by Basso, Farhi, Marwaha, Villalonga and Zhou (2022), §7, equation (7.2).

The circuit proof needs a finite correlation construction as well as the scalar equality. Fix $0 < T < 1$, and let N denote the mesh count, distinct from the number of spins n. OpenAI (2026a), Proposition 7.2 constructs, for all sufficiently large N, from independent standard normals $Z_1, \dots, Z_{N+1}$, a sign-valued function F_N and normalised increments $g_j = h_j(Z_1, \dots, Z_{j-1})Z_j$, $1 \le j \le N$, with $\mathbb{E}g_j^2 = 1$, such that

$$\lim_{N\to\infty} \sum_{j=1}^{N} \mathbb{E}[F_N g_j]\, \mathbb{E}[F_N Z_{j+1}] = \int_0^T \mathbb{E}[a(t, X_t)]\, dt. \tag{11}$$

The multipliers h_j are normalised Hessians along the Euler approximation to X. The extra Gaussian rounds the terminal gradient to a sign while preserving its conditional mean and all preceding correlations. Both correlation vectors have bounded ℓ^2 norm; their ℓ^2 approximation errors vanish, so pairing preserves the limit. The products with $j < N$ give a Riemann sum, while the final rounding-coordinate term vanishes. The mesh limit is taken at fixed T; only then does $T \uparrow 1$ give P_*. The construction and its error estimates are in long §7, "A finite Gaussian spin and its adjacent correlations".

The adjacent index in (11) is determined by the energy in the rooted Gaussian construction of OpenAI (2026b), Lemma 3.1. A normalised child direction g_j generates the parent coordinate Z_{j+1}. Projection of F_N onto the linear Gaussian coordinates (the first Gaussian chaos) gives coefficient $\mathbb{E}[F_N Z_{j+1}]$ at that coordinate. The rooted energy pairs this coefficient with the coefficient of its generating child direction, $\mathbb{E}[F_N g_j]$. Their product is therefore the jth contribution in (11). The initial independent root seed contributes zero. This identifies the energy as well as the Gaussian law. The recursive construction and projection are proved in long §8.1.

The remaining circuit steps preserve the identified energy to a chosen tolerance. Regularisation puts the finite formula in the class used by the synthesis theorem, which realises it with temporary Gaussian seed controls in addition to cost and mixer gates. The same sitewise seeds are reused throughout that word. A further theorem simulates their joint effect using only cost and mixer gates; only then does fixed-word transfer identify the thermodynamic SK value. These are separate results of OpenAI (2026b), §§2–6, reconstructed in long §§8.1–8.6, "From the scalar formula to one ordinary QAOA word". The source's Appendix A supplies an alternative positive-temperature input to the same synthesis. The selected argument gives no quantitative depth bound, finite-size rate or efficient procedure for finding the angles.

6. Relation to the formal statements

This AI-assisted article reconstructs the cited analytic arguments. The integrated estimate (5), its shape hypotheses and compact-strip approximation are substantial imported inputs.

The full-support formal statement asserts the support and moment conclusions conditional on an admissible minimiser, and constructs its associated diffusion. A separate scalar formal statement covers value identities and finite Gaussian approximation, with minimiser and diffusion premises for those clauses. Its third clause identifies the ground-state and Parisi values from a minimiser, constructing the diffusion internally. The two formal models have not been transported into one

another in this record. Neither statement includes the positive-density corollary or QAOA convergence. Their full-support and scalar implementations use Lean and mathlib.

No completed research-target acceptance by Comparator, the checker comparing an implementation with its formal challenge, is recorded for this edition. Exact versions, checking outcomes and mathematical correspondence are retained in *Source review for the QAOA–SK drafts*, edition 25 October 2026, and the long record's §12, "What the formal statements establish", and §13, "Source checks and finite calculation".

References

- de Moura, L. and Ullrich, S. (2021). "The Lean 4 Theorem Prover and Programming Language (System Description)." *CADE 28*. System description.
- The mathlib Community (2020). "The Lean Mathematical Library." *Certified Programs and Proofs (CPP '20)*. Library description.
- OpenAI (2026a). *Full support of the zero-temperature Sherrington–Kirkpatrick order parameter*. 27 September 2026. Manuscript, fixed revision. Theorem 1.1; §§2–6; Corollaries 6.4 and 7.1; Proposition 7.2.
- OpenAI (2026b). *QAOA attains the SK ground-state energy in the thermodynamic-first limit*. 25 September 2026. Manuscript, fixed revision. Theorem 1.1; §§2–6 and Appendix A.
- Auffinger, A. and Chen, W.-K. (2017). "Parisi formula for the ground state energy in the mixed p-spin model." *Annals of Probability* **45**, 4617–4631. Preprint v2: Theorem 1 and proof of Lemma 3.
- Auffinger, A., Chen, W.-K. and Zeng, Q. (2020). "The SK model is infinite step replica symmetry breaking at zero temperature." *Communications on Pure and Applied Mathematics* **73**, 921–943. Preprint v2: Theorem 1 and Remark 7.
- Chen, H.-B. (2026). *FRSB in the SK spin glass: convergence to full-interval support at zero temperature*. Version 1, 20 July 2026. §1, p. 4; Theorem 1.3; Propositions 4.2, 4.5 and 4.6; Appendices B.6 and C.
- Lopatto, P. (2026). *Full replica symmetry breaking in the Sherrington–Kirkpatrick model*. arXiv version 3, revised 16 July 2026. Propositions 3.6, 7.1 and 9.1; related Cole–Hopf, density and crossing arguments. The separate version 4, Proposition 5.1 and Corollary 5.6, supplies the later comparison.
- Lopatto, P. (2026). *Parisi measure of the Sherrington–Kirkpatrick model with external field*. arXiv version 1. Theorem 2.3 and Appendix B; related positive-temperature two-use argument.
- Sellke, M. (2021). *Optimizing Mean Field Spin Glasses with External Field*. Preprint v4, 6 October 2021. Proposition 3.10 and Lemma 3.11, p. 17.
- Sherrington, D. and Kirkpatrick, S. (1975). "Solvable Model of a Spin-Glass." *Physical Review Letters* **35**, 1792–1796. Original model.
- Jagannath, A. and Tobasco, I. (2016). "A dynamic programming approach to the Parisi functional." *Proceedings of the AMS* **144**, 3135–3150. Preprint v4. Positive-temperature control methods.
- Chen, W.-K., Handschy, M. and Lerman, G. (2018). "On the energy landscape of the mixed even p-spin model." *Probability Theory and Related Fields* **171**, 53–95. Preprint v4: Proposition 2, Theorem 5, Lemma 3 and Propositions 3–4.
- Auffinger, A. and Chen, W.-K. (2015). "On properties of Parisi measures." *Probability Theory and Related Fields* **161**, 817–850. Preprint v1: Lemma 2 and proof of Theorem 2(ii).

- Montanari, A. (2025). "Optimization of the Sherrington–Kirkpatrick Hamiltonian." *SIAM Journal on Computing* **54**, FOCS19-1–FOCS19-38. Preprint v2, 2019: §§2–3.
- El Alaoui, A., Montanari, A. and Sellke, M. (2021). "Optimization of mean-field spin glasses." *Annals of Probability* **49**, 2922–2960. Preprint v1: Theorem 2, p. 6; Remark 2.1 and Theorem 3, p. 7; Assumption 2 and Corollary 2.2, p. 8; §§5–6.
- Farhi, E., Goldstone, J. and Gutmann, S. (2014). *A Quantum Approximate Optimization Algorithm*. Preprint v1. Algorithm definition; §VI concerns the fixed-instance depth limit.
- Farhi, E., Goldstone, J., Gutmann, S. and Zhou, L. (2022). "The Quantum Approximate Optimization Algorithm and the Sherrington–Kirkpatrick Model at Infinite Size." *Quantum* **6**, 759. Preprint v4.
- Basso, J., Farhi, E., Marwaha, K., Villalonga, B. and Zhou, L. (2022). "The Quantum Approximate Optimization Algorithm at High Depth for MaxCut on Large-Girth Regular Graphs and the Sherrington–Kirkpatrick Model." *TQC 2022*, 7:1–7:21. Full preprint v3: transfer theorem and §7, equation (7.2); the full preprint has separate numbering.